Singapore International School @ DaNang

School Year 2025-2026

COMPUTER SCIENCE – SEMESTER 2 EXAM

Name of Candidate:			
Class Teacher:	MR GARY	Subject:	COMPUTER SCIENCE
Year Level:	IG1	Duration:	1 HOUR
Term:	4	**Exam Date:**	

Instructions	• Please write your name and class on this sheet. • Please answer all questions in the sheet provided. • Please write neatly, especially if you write more than the allotted space.
Exam Materials	• Paper 2

Results:	______/	**Grade:**
Comments:		
Parent's Name & Signature		

Relevant Expected School Wide Learning Goals	
Singapore International School seeks to nurture ACADEMIC ACHIEVERS who:	**Singapore International School seeks to nurture CRITICAL THINKERS who:**
• Meet or exceed year level expectations • Strive for continuous improvement	• Work collaboratively with others • Apply rational, higher order thinking skills

Algorithm Design and Problem Solving

1. Identify **two** specific tasks that a programmer would perform during the **analysis** stage of the program development life cycle. [2]

____

____

2. A program requires a user to enter their age, which must be between 11 and 18 inclusive. [3]

a. State the most appropriate **validation check** for this input.

____

b. Provide one example of **abnormal (erroneous) test data** for this input.

____

c. Provide one example of **extreme test data** for this input.

____

3. Complete the trace table for the following algorithm using the test data: **12, 45, 8, 20**. [4]

```
Total ← 0
Counter ← 0
REPEAT
  INPUT Number
  IF Number > 10
  THEN
    Total ← Total + Number
  ENDIF
  Counter ← Counter + 1
UNTIL Counter = 4
OUTPUT Total
```

Number	Total	Counter	OUTPUT
	0	0	

4. Write a **pseudocode algorithm** that meets the following requirements. [10]

- The program must allow a user to enter the **weight** of 30 items.
- Each weight must be **validated** to ensure it is greater than 0 and less than or equal to 100.
- The program must calculate and output the **average weight** of all items.
- The program must **count** and output how many items weighed more than 50.
- You must use **meaningful variable names** and appropriate **initialisation**.

5. Draw a structure diagram for a "Pizza Ordering System." [2]

- The system should be decomposed into three main sub-systems: "Input Order," "Process Payment," and "Output Receipt."
- Decompose the "Input Order" sub-system into two further sub-systems: "Select Size" and "Choose Toppings."

6. a. Define verification as it relates to data entry. [2]

____

____

____

6. b. Identify and describe one method of verification for input data. [2]

____

____

____

7. An algorithm was intended to find the highest number in a list of 10 positive integers, but it contains errors.

```
01 Max ← 100
02 Counter ← 0
03  WHILE Counter < 10
04      INPUT Number
05      IF Number < Max
06      THEN
07      Max ← Number
08      ENDIF
09      Counter ← Counter + 1
10  ENDWHILE
11 OUTPUT Max
```

a. Identify the two line numbers containing errors. [2]

1.

2.

b. Describe how to correct these errors so the algorithm correctly finds the highest number. [4]

1. ____

____

____

2. ____

____

____

8. Explain the steps of a linear search algorithm. Describe how a linear search would find the value **"Apple"** in an array called FruitList containing 50 items. [4]

9. Identify the purpose of the following pseudocode algorithm.

```
INPUT SearchValue
Found ← FALSE
Index ← 1
WHILE Found = FALSE AND Index <= 100 DO
  IF MyArray[Index] = SearchValue
  THEN
    Found ← TRUE
  ELSE
    Index ← Index + 1
  ENDIF
ENDWHILE
IF Found = TRUE
THEN
  OUTPUT "Value located at ", Index
ELSE
  OUTPUT "Not in list"
ENDIF
```

a. State the **purpose** of the algorithm. [1]

b. Describe **two processes** (actions) that occur within this algorithm. [4]

1.

2.